Information Theory Mn908 and MICAS903

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Solutions are due on November 5, 2026, at 9h00.

Problem 1 The Z-Channel

The Z-channel has binary input and output alphabets and a transition probability $W(\cdot \mid \cdot)$ given by

$$W = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad X, Y \in \{0, 1\}.$$

(The channel is called a Z-channel because its channel diagram looks like the letter Z.) Find the capacity of the Z-channel and the maximizing input probability distribution.

Problem 2 Cascade Channel

1. Show that the cascade channel in Figure **??**, from input X to output Y, has the same transition law as a BSC($d/2$). By a cascade channel we mean that the output of the first channel is directly fed as input to the second channel. Thus, Z is both the output of the first channel and the input to the second channel.
2. Show that the capacity of a BSC($d/2$) cannot be larger than the capacity of a BEC(d).

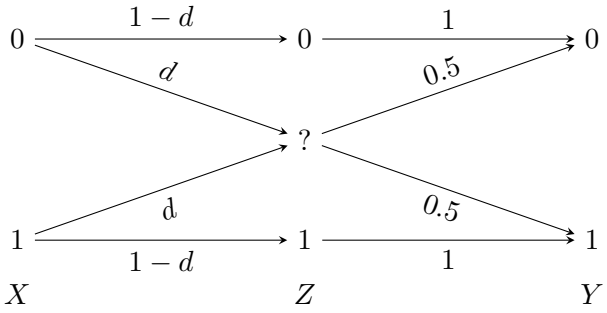


Figure 1: Cascade channel.

Problem 3 Capacity of a Sum Channel

1. Consider n discrete memoryless channels

   $$\{(\mathcal{X}_i, \mathcal{Y}_i, P_{Y_i|X_i})\}_{i=1}^n$$

   with capacities $C_1, C_2, \ldots, C_n$. We assume that the input alphabets $\{\mathcal{X}_i\}_{i=1}^n$ are disjoint and the output alphabets $\{\mathcal{Y}_i\}_{i=1}^n$ are disjoint.

   The sum channel is defined as the union of these channels, where at each time only one of the channels can be used. Thus, the sum channel has input alphabet

   $$\mathcal{X} = \bigcup_{i=1}^n \mathcal{X}_i,$$

output alphabet

$$\mathcal{Y} = \bigcup_{i=1}^{n} \mathcal{Y}_i,$$

and transition probability

$$P_{Y|X}(y \mid x) = \begin{cases} P_{Y_i|X_i}(y \mid x), & y \in \mathcal{Y}_i,\ x \in \mathcal{X}_i \text{ for some } i, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Show that the capacity of the sum channel is

$$C = \log_2\left(\sum_{i=1}^{n} 2^{C_i}\right),$$

and find P_X, the capacity-achieving input probability distribution. Also interpret C as the average of the capacities of the individual channels plus the information conveyed by the selection of a channel.

Hint: Let S denote a random variable that takes the value i if the input X is in $\mathcal{X}_i$. Thus, S determines the choice of channel. Show first that

$$I(X;Y) \le H(S) + \sum_{i=1}^{n} q_i C_i, \tag{2}$$

where $q_i \triangleq \Pr[S = i]$, and identify when the inequality holds with equality. Then write the right-hand side of (2) as the difference of a constant and a Kullback–Leibler divergence that depends on $q_1, \ldots, q_n$, and find the maximizing $q_1, \ldots, q_n$.

2. Use the above result to find the capacity of a channel with $\mathcal{X} = \mathcal{Y} = \{0, 1, 2\}$ and

$$P_{Y|X}(2 \mid 2) = 1,$$

$$P_{Y|X}(0 \mid 0) = P_{Y|X}(1 \mid 1) = 1 - \epsilon,$$

and

$$P_{Y|X}(1 \mid 0) = P_{Y|X}(0 \mid 1) = \epsilon, \qquad \epsilon \in (0, 1/2).$$

Problem 4 The Binary Jammer Channel

Consider the problem of communicating over a binary symmetric channel with random interference. For each input bit, there is a probability q that the channel is "free," in which case it behaves as an ordinary binary symmetric channel with crossover probability ϵ. With probability $1 - q$, however, the channel is "blocked" by an interferer. When the channel is blocked, the interferer replaces your input bit with a random input bit drawn according to $p(0) = p(1) = 1/2$. Find the capacity C of the jammer channel under the following conditions:

1. Neither the encoder nor the decoder knows whether the channel is blocked at each time.
2. The decoder knows when the channel is blocked, but the encoder does not.
3. Show that if both encoder and decoder know the channel state, the rate

$$q(1 - H_{\mathrm{b}}(\epsilon))$$

is achievable by demonstrating a coding scheme that achieves this rate.